

Multiagent Federated Reinforcement Learning for Secure Incentive Mechanism in Intelligent Cyber–Physical Systems

Minrui Xu¹, Jialiang Peng², B. B. Gupta³, *Senior Member, IEEE*, Jiawen Kang⁴, Zehui Xiong,
Zhenni Li⁵, *Member, IEEE*, and Ahmed A. Abd El-Latif⁶, *Senior Member, IEEE*

Abstract—Federated learning (FL) is an emerging technology for empowering various applications that generate large amounts of data in intelligent cyber–physical systems (ICPS). Though FL can address users’ concerns about data privacy, its maintenance still depends on efficient incentive mechanisms. For long-term incentivization to participants in data federation under dynamic environments, deep reinforcement learning as a promising technology has been extensively studied. However, the nonstationary problem caused by the heterogeneity of ICPS devices results in a serious effect on the convergence rate of existing single-agent reinforcement learning. In this article, we propose a multiagent learning-based incentive mechanism to capture the stationarity approximation in FL with heterogeneous ICPS. First, we formulate the secure communication and data resource allocation problem as a Stackelberg game in FL with multiple participants. Then, to tackle the heterogeneous problem, we model this multiagent game as a partially observable Markov decision process. In particular, a multiagent federated reinforcement

learning algorithm is proposed to learn the allocation policies efficiently by dwindling variances in policy evaluation caused by interaction among multiple devices without the requirement of sharing privacy information. Moreover, the proposed algorithm is proved to attain convergence at an expected rate. Finally, extensive experimental results demonstrate that our proposed algorithm significantly outperforms baseline approaches.

Index Terms—Deep reinforcement learning (DRL), federated learning (FL), intelligent cyber–physical systems (ICPS).

I. INTRODUCTION

WITH the rapid development of intelligent technologies, cyber–physical systems [1], [2] have evolved into intelligent cyber–physical systems (ICPS), which has significantly promoted the popularity of a multitude of emerging intelligent applications in our lives, such as smart agriculture, robotic systems, and intelligent transportation. These data-driven intelligent applications generate an enormous amount of data. However, this data is confined within edge devices due to limited bandwidth and reliability of wireless networks, and thus hinders the development of intelligent services in ICPS [3]–[6]. Thanks to the advance of mobile-edge computing (MEC) [7]–[9], local data storing and processing are possible in edge devices, such as drones, vehicles, and roadside units, which are equipped with storage and computation capability. Noticing data owners may be unwilling to share their private raw data with competitors [10]–[12], a new distributed machine learning technology named federated learning (FL) enables these distributed devices to build up intelligent services collaboratively without exposing their private data.

Nonetheless, motivating heterogeneous devices within ICPS to participate in FL by contributing their data and computation resources is a nontrivial task even with the great advantages in FL. This crucial challenge lies in the incentive mechanism for resource trading of FL, in terms of computation, network, and data resources [13]–[17]. Recent developments in deep reinforcement learning (DRL) provide promising approaches [18]–[20] to improve efficiency and quality of service in MEC purely based on past experience, without knowing any prior information about systems. For example, the deep Q network is used to approximate the Q value-action function to improve the network’s sum rate and energy

Manuscript received 12 January 2021; revised 15 April 2021; accepted 3 May 2021. Date of publication 18 May 2021; date of current version 7 November 2022. This work was supported in part by the National Key Research and Development Plan under Grant 2018YFB1003803; in part by the National Natural Science Foundation of China under Grant 61803096, Grant 62073086, and Grant 61802450; in part by the Natural Science Foundation of Guangdong under Grant 2018A030313005; in part by the Pearl River Talent Recruitment Program under Grant 2019QN01X130; in part by the Program for Guangdong Introducing Innovative and Entrepreneurial Teams under Grant 2017ZT07X355; in part by the Guangzhou Science and Technology Program Project under Grant 202002030289; and in part by the Fundamental Research Funds for Heilongjiang Universities, China under Grant 2020-KYYWF-1014. (Corresponding author: Zhenni Li.)

Minrui Xu is with the School of Computer Science and Engineering, Sun Yat-sen University, Guangzhou 510006, China (e-mail: xumr3@mail2.sysu.edu.cn).

Jialiang Peng is with the School of Data Science and Technology, Heilongjiang University, Harbin 150080, China (e-mail: jialiangpeng@hlju.edu.cn).

B. B. Gupta is with the Department of Computer Engineering, National Institute of Technology Kurukshetra, Kurukshetra 136119, India, and also with the Department of Computer Science and Information Engineering, Asia University, Taichung City 413, Taiwan (e-mail: gupta.brij@gmail.com).

Jiawen Kang is with the Joint NTU-WeBank Research Centre on Fintech, Nanyang Technological University, Singapore 639798 (e-mail: kavinkang@ntu.edu.sg).

Zehui Xiong is with the Pillar of Information Systems Technology and Design, Singapore University of Technology and Design, Singapore 487372 (e-mail: zehui_xiong@sutd.edu.sg).

Zhenni Li is with the School of Automation and the Guangdong Key Laboratory of IoT Information Technology, Guangdong University of Technology, Guangzhou 510006, China (e-mail: lizhenni2012@gmail.com).

Ahmed A. Abd El-Latif is with the Department of Mathematics and Computer Science, Faculty of Science, Menoufia University, Shebin El-Koom 32511, Egypt (e-mail: a.rahim@gmail.com).

Digital Object Identifier 10.1109/IIOT.2021.3081626

efficiency under time-varying wireless network [21]. In the context of FL, which is naturally a multiagent system consisting of a variety of heterogeneous devices [22], multiagent reinforcement learning (MARL) is proposed to address non-stationary problems caused by interaction among multiple devices [23]. Specifically, MARL requires intelligent learning agents to exchange their observation or share experience to learn to communicate or cooperate with other agents [24]. By coordinating among these agents, MARL dwindles variance in the evaluation of their policies, thus accelerates their convergence rate.

However, coordination among these agents may cause serious security and privacy issues if we directly exploit existing MARL algorithms in privacy-preserving ICPS [25], [26]. In detail, agents have privacy concerns on their privacy information and are reluctant to share their observation and experience with other agents. For example, in most industrial ICPS [27], experience replays of agents exist in the form of isolated islands [28]. Due to industry competition and privacy security, it is difficult for agents to integrate data between different departments of the same factory faces heavy resistance. In this way, agents can only gather scanty experience for policy iteration, which leads to a high variance of policy performances in policy evaluation, thus resulting in an unsatisfactory convergence rate of policies.

To address these issues, it is natural to require agents to coordinate with other devices to develop policies collaboratively without sharing privacy information [24]. In this article, we proposed a multiagent federated reinforcement learning (MA-FRL) algorithm where agents can share their local policies in a secure-enhanced manner instead of directly sharing observation or experience, which may contain sensitive information. Specifically, there are three main procedures in the proposed algorithm. First, agents of MA-FRL use their local experience replays to evaluate and improve their policies, which converge to their local experience. Then, agents encrypt their policies utilizing privacy protection techniques, such as homomorphic encryption, secret sharing, and differential privacy, to encrypt these policies, before uploading them to aggregators, in case some malicious participants drive any privacy information from parameters of these policies [16]. Finally, aggregators synthesize these policies into global policies using the FedAvg [29] algorithm and then distribute these global policies to agents, which contain lower variances in policy evaluation. By repeating these procedures, agents can finally get aggregated policies with stronger inferential ability in a satisfactory convergence rate.

In contrast with previous studies, we design the MA-FRL algorithm, which is privacy preserving, to leverage a novel training approach for multiagent systems, where multiple intelligent agents collaboratively learn global policies without sharing privacy information. In this way, the convergence rate of MARL is remarkably accelerated due to a better balance between exploration and exploitation during the training process. It should be noted that the MA-FRL algorithm is designed to tackle the aforementioned challenges and is not limited by incentive mechanism scenarios. It can be applied to many problems that need to derive the stationary assumption

without sharing privacy information in any multiagent systems.

Our main contributions can be summarized as follows.

- 1) We study and formulate the secure resource allocation problem in multiagent FL as a Stackelberg game. To address the heterogeneity brought by multiple agents, this game is formulated as a partially observable Markov decision process (POMDP).
- 2) To tackle the nonstationary problem in FL with various devices, we propose the MA-FRL algorithm which can synthesize a good inferential global policy from encrypted local policies of multiple agents without shared privacy information. Additionally, we derive the expected convergence rate of the proposed algorithm.
- 3) To demonstrate that the proposed algorithm matches the theoretical analysis and outperforms existing approaches, we implement the MA-FRL algorithm on both real-world and simulating experiments.

The remainder of this article is outlined as follows. We discuss the related work in Section II. We present the system model in Section III and the problem is formulated in Section IV. We then present the design and implementation details of the proposed MA-FRL algorithm in Section V. We simulate our result in Section VI. Finally, we conclude this article in Section VII.

II. RELATED WORKS

A. Federated Learning for Intelligent CPS

FL over wireless networks has been studied extensively [30], [31]. Owing to the uncertainty of wireless channels, there are various tradeoffs existing in FL, such as between the FL time and energy consumption of ICPS devices. For example, Zhang *et al.* [4] have proposed an FL-based edge computing platform for ICPS, named “FengHuoLun,” which ensures the development of trustworthy smart services given the dynamic and distributed nature of ICPS. Furthermore, a secure and intelligent architecture in vehicular ICPS, which involves multiple users and multiple transmission channels, has been proposed by Lu *et al.* [3] for enhancing data privacy. Withstanding cyber threats to large-scale threats against industrial ICPS, Li *et al.* [32] have designed a new deep-learning-based intrusion detection model for industrial ICPS, by making use of a convolutional neural network and a gated recurrent unit. In real-world ICPS, diverse computing power, heterogeneous tasks, and decentralized storage and acquisition of training data are still limitations. Therefore, Wang *et al.* [2] have proposed an online federated knowledge distillation method as heterogeneous brain storming to tackle these issues.

B. DRL-Based Algorithms for Federated Learning

With the advent of DRL [33]–[36], the above issues can be addressed by formulating the FL in ICPS as a Markov decision process and learning the long-term policy from the interaction between the agent and the environment. For example, due to the unique challenges of unshared decisions and difficulties

of contribution evaluation, Zhan *et al.* [37] designed a DRL-based incentive mechanism to determine the optimal pricing strategy for the parameter server (PS) and optimal training strategies for edge nodes (ENs). Zhan and Zhang [38] have used DRL to learn a pricing strategy purely based on the experience, which can capture the tradeoff between payment and training time. A DRL-based experience-driven algorithm has been proposed by Zhan *et al.* [39] to improve the energy efficiency and minimize the total system cost. In FL under ultradense edge computing, a two-timescale DRL approach has been designed by Yu *et al.* [40] to achieve real-time and low-overhead resource allocation strategies. Nevertheless, these single-agent algorithms are not practical for nonstationary ICPS environments, where heterogeneous devices frequently interact with other devices.

To better overcome these obstacles, we need to take the advantages of the MARL, which allows agents to develop policies collaboratively regarding the whole environment [41]. For instance, Kwon *et al.* [42] have proposed a joint cell association and resource allocation method, basing on multiagent deep deterministic policy gradient, for smart ocean FL. While MARL can restore the Markov property by learning communication or cooperation, increasing privacy concerns shed light on privacy-preserving MARL, i.e., FRL. For example, Messaoud *et al.* [43] have proposed a federated deep Q -learning algorithm for dynamic network management and resource allocation of heterogeneous quality-of-service applications in Industrial Internet of Things. For smart homes, Lee and Choi [44] have proposed an FRL approach to optimize energy management home appliances, including a solar photovoltaic system and an energy storage system. Wang *et al.* [45] have designed an attention-weighted federated DRL to optimize the node selection and cache replacement for device-to-device-assisted heterogeneous collaborative edge caching framework in mobile networks. However, these presented algorithms are designed for other privacy-preserving environments and cannot directly be applied to improve the performance of secure communication and resource allocation in collaborative edge learning. This work, to the best of our knowledge, is the first attempt to exploit privacy-preserving MARL for FL.

III. SYSTEM MODEL

As illustrated in Fig. 1, we consider a cooperative nonorthogonal multiple access (C-NOMA) wireless multiuser systems consisting of one PS and a set $\mathcal{N}$ of N end devices mentioned in [46]. These end devices are distributed in a set $\mathcal{B}$ of B resource blocks and connect to the PS through the ENs. Inside resource blocks, there are near devices (NDs) located close to ENs, far devices (FDs) located far from ENs, and relay devices (RDs). Grouping one ND, one FD, and one RD in each resource block is an effective and practical implementation for secure communication [47], [48], where FDs in global model downloading and NDs in local model uploading adaptively emit jamming signals to untrusted RDs to impair their eavesdropping capability. Meanwhile, the size of the model updates is not affected because RDs will combine its

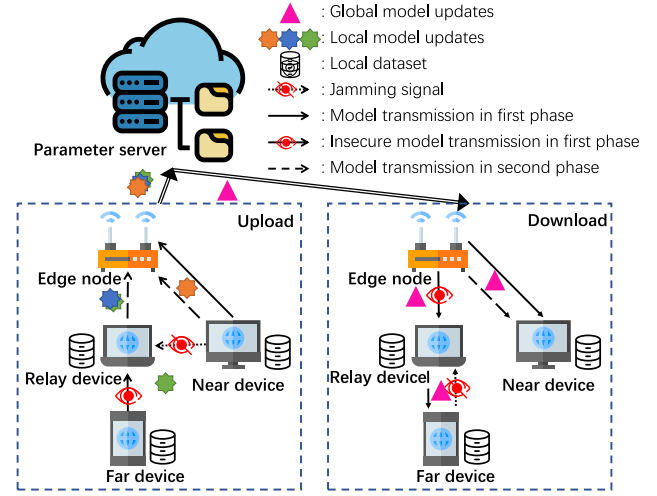


Fig. 1. Secure FL involving upload block and download block.

model update with its received model update with the average operation [49].

Each participating device n stores its private data set $\mathcal{Q}_n$ of size Q_n , respectively. $\mathcal{Q}_n$ defines the collection of data samples given as a set of input-output pairs $\{x_i, y_i\}_{i=1}^{Q_n}$, where $x_i \in \mathbb{R}^d$ is an input sample vector with d features, and y_i is the labeled output value for the sample x_i . Denote $f_i(\mathbf{w})$ as the loss function on the data sample i . Some examples of the loss function are $f_i(\mathbf{w}) = (1/2)(x_i^T \mathbf{w} - y_i)^2$, $y_i \in \mathbb{R}$ for linear regression and $f_i(\mathbf{w}) = \{0, 1 - y_i x_i^T \mathbf{w}\}$, $y_i \in \{-1, 1\}$ for support vector machine. Each participating n uses its data set $\mathcal{Q}_n$ to train a local model $\mathbf{w}_i$ and sends only the local model to the parameters after homomorphic encryption, secret sharing, and differential privacy. Therefore, the total size of data samples from N end device is $\sum_{n=1}^N Q_n = Q$. To describe the article more clearly, we list the important notations used in this article in Table I.

A. Federated Learning Objective

The objective for FL is to minimize the overall loss with respect to (w.r.t.) the local data set of participants. And for each local data set $\mathcal{Q}_n$ at end device n , the loss function is defined as

$$F_n(\mathbf{w}) := \frac{1}{Q_n} \sum_{i \in \mathcal{Q}_n} f_i(\mathbf{w}) + \zeta g(\mathbf{w}) \quad (1)$$

where $\mathbf{w} \in \mathbb{R}^d$ is the parameters of the local model, and $g(\cdot)$ is a regularizer function, commonly expressed as $g(\cdot) = (1/2)\|\cdot\|^2$; $\forall \zeta \in [0, 1]$. This characterizes the local model in FL setting. Then, the learning model is a minimizer of the following global loss function minimization problem:

$$\min_{\mathbf{w} \in \mathbb{R}^d} F(\mathbf{w}) := \sum_{n=1}^N \frac{Q_n}{Q} F_n(\mathbf{w}). \quad (2)$$

B. Secure Federated Learning

In general, the FL training process includes the following four steps. Note that the local models refer to the trained models at participating devices, whereas the global model refers to the model aggregated by ENs.

TABLE I
LIST OF NOTATIONS AND DEFINITIONS

Notion	Definition
$\mathcal{N}$	Set of ICPS devices
$\mathcal{B}$	Set of resource blocks
$\mathcal{Q}_n$	Private dataset of device n
F_n	Loss function of device n
$\mathbf{w}_n$	Local model of device n
$C_{b,n}, C_{b,r}, C_{b,f}$	Cooperative transmission rate of ND, RD, FD
l_p	Leak probability
τ	Number of local iteration of device n
θ_n	Relative local model performance of device n
ϕ_n	Data quality in device n
Q_n	Data quantity in device n
u_k	Utility of device n
p_n	Price given to device n
U	Training cost
T_{global}	Total training time
K	Bound improvement ration
S_b^k	State space in resource block b at round k
$a_{b,s}^k, a_{b,n}^k, a_{b,r}^k, a_{b,f}^k$	Action of EN, ND, RD, FD
$o_{b,s}^k, o_{b,n}^k, o_{b,r}^k, o_{b,f}^k$	Observation of EN, ND, RD, FD
ϑ_b^e	Policy in resource block b at episode e
L_k^A	Policy loss w.r.t experience from round k

1) *Global Model Download*: At the beginning of each global iteration, ENs distribute the global model to participants, as illustrated in the right side of Fig. 1 and the collaborative model download consists of two phases within each fading block. In the first phase, ENs transmit the global model to NDs and RDs. In order to prevent the leakage of information at RDs, FDs are designated to transmit a portion of their transmit power for emitting jamming signals along with ENs' transmission. During the second phase, RDs forward their received model to FDs. If RDs can decode superimposed code from ENs faster than FDs, there would be a risk of privacy information leaking.

2) *Local Update*: The local update in each iteration is performed on the parameters after possible global aggregation in the previous iteration. For each device n , the update rule is as follows:

$$\mathbf{w}_n^{k+1} = \mathbf{w}^k - \eta \nabla F_n(\mathbf{w}^k) \quad (3)$$

where $\eta \geq 0$ is the stepsize. Depending on computation capacities of devices, they can repeat this steps for better local models.

3) *Local Model Upload*: For the RDs in the system, they not only need to train their local models but also aggregate the local models received from other devices. As depicted in Fig. 1, during the first phase, ENs receive the individual model from NDs and RDs receive individual models from FDs. To prevent RDs from intercepting the conveyed signals intended for ENs, during the first phase, jamming signals are sent by NDs with a portion of their transmit power. During the second phase, RDs transmit the aggregated model to ENs. If RDs can decode superimposed code from FDs faster than ENs, there would be a risk of privacy information leaking.

4) *Global Aggregation*: At the PS, the following global model is aggregated:

$$\mathbf{w}^{k+1} = \frac{\sum_{n=1}^N \mathcal{Q}_n \mathbf{w}_n^{k+1}}{Q}. \quad (4)$$

In this way, distributed participants can train a global model collaboratively. In the next section, we will formulate this problem as a game to motivate heterogeneous devices to join the learning.

IV. PROBLEM FORMULATION

In this section, we formulated the secure resource allocation problem in FL as a Stackelberg game, where the ENs are the leaders of this game and end devices are the followers. We first put forward the secure cooperative communication model. Then, we put forward the model evaluation method that reflects the marginal revenue of multiple local iterations. We formulate the utility function of end devices and analyze the optimal utility corresponding to the price. Finally, the utility function of ENs is formulated as an optimization problem minimizing the long-term training cost, in terms of expected payment.

A. Secure Communication Model

As the aforementioned secrecy-enhancing model download and upload, we then give the specific communication model. Considering one resource block, the channel gains of links between EN s and ND n , between EN s and RD r , between RD r and ND n , and between RD r and FD f are denoted as $g_{s,n}, g_{s,r}, g_{r,n}$, and $g_{r,f}$, respectively. The average transmit power at each device is upper bounded by P , for both download and upload phases. Denoting the additive white Gaussian noise with n_0 , the system signal-to-noise ratio is denoted by $\rho = (P/n_0)$. During the transmit power allocation phase, ρ_n, ρ_f , and ρ_r denote the C-NOMA power allocation coefficients at the same resource block with $\rho_n + \rho_f + \rho_r \leq \rho$.

There are two phases in cooperative communication. In the first phase, the transmission rate $y_{n,s}, y_{f,r}$, and $y_{n,r}$ w.r.t. the system bandwidth W between ND n and EN s , FD f and RD r , and ND n and RD r , can be computed as

$$y_{n,s} = \frac{1}{2} W \log(1 + \Gamma_{n,s}) \quad (5)$$

$$y_{f,r} = \frac{1}{2} W \log(1 + \Gamma_{f,r}) \quad (6)$$

$$y_{n,r} = \frac{1}{2} W \log(1 + \Gamma_{n,r}) \quad (7)$$

where $\Gamma_{n,s} = \Omega \rho_n g_{r,f}$, $\Gamma_{f,r} = [(\rho_f g_{r,f})/(\rho_n g_{r,n} + 1)]$, and $\Gamma_{n,r} = [(\Omega \rho_n g_{r,n})/(1 - \Omega) \rho_n g_{r,n} + 1]$. Here, the jamming power proportion Ω is adaptively allocated by FDs in the download phase and NDs in the upload phase.

In the second phase, the transmission rate $y'_{n,s}$ and $y_{r,s}$ w.r.t. amplifying coefficient $G = [(\rho_r)/(\rho_n g_{r,n} + \rho_f g_{r,f} + 1)]$ between ND n and EN s , RD r and EN s can be calculated as

$$y'_{n,s} = \frac{1}{2} W \log(1 + \Gamma'_{n,s}) \quad (8)$$

$$y_{r,s} = \frac{1}{2} W \log(1 + \Gamma_{r,s}) \quad (9)$$

where $\Gamma'_{n,s} = [(\rho_{ngs,n})/(G\rho_{rgs,rg}(r,f) + Gg_{s,r} + 1)]$ and $\Gamma_{r,s} = [(\rho_{rgs,rg}(r,f))/g_{s,r} + (1/G)]$. As a result, the cooperative transmission rate for device n, f , and r can be calculated as

$$C_{b,n} = \{y_{n,s} - y_{n,r}\}^+ + y'_{n,s} \quad (10)$$

$$C_{b,f} = \{y_{r,s} - y_{f,r}\}^+ \quad (11)$$

$$C_{b,r} = \{y_{r,s} - C_{b,f}\}^+ \quad (12)$$

within resource block b . During the upload period, if $y_{n,s} > y_{n,r}$, then there would be a risk for privacy information being eavesdropped and $l(k) = 1$. Besides, if $y_{r,s} > y_{f,r}$ during the model download period, there would be also risky to be eavesdropped and $l(k) = 1$. In this way, the leak probability l_p within one episode can be denoted as

$$l_p = \frac{\sum_{k=0}^K l(k)}{K} \quad (13)$$

where K is the length of the episode. After the secure communication model defined above, we then give the evaluation for the training model.

B. Model Evaluation

Let τ_n denote the number of local iterations at the device n between every two global aggregations. Similar to [13], [16], and [50], the relative local model performance then satisfies

$$\theta_n(\tau_n^k) = e^{-\psi(\phi_n Q_n \tau_n^{k|\mathcal{A}|})^v} \quad (14)$$

where ϕ_n denotes the quality of data in device n and Q_n denotes the quantity of data in device n . ψ and v are weight factors. $|\mathcal{A}|$ is the learning factor to reflect the marginal revenue of multiple local iterations depending on the chosen learning algorithm $\mathcal{A}$ and the step size η in (3). To characterize the quality of the local solution, the local relative θ_n^k accuracy satisfied in (14), in each round, k also satisfies

$$\mathbb{E}[F_n(\mathbf{w}_n^*) - F_n(\mathbf{w}_n^k)] \leq \theta_n^k [F_n(\mathbf{w}_n^*) - F_n(\mathbf{0})] \quad (15)$$

where $F_n(\cdot)$ denotes the local loss function defined in (1), $\mathbf{0}$ denotes the initial model parameter, and $\mathbf{w}_n^*$ is the optimal model parameter for device n . Giving the two extreme examples, achieving $\theta_n^k = 0$ would require to find the exact maximum, while $\theta_n^k = 1$ states that no improvement was achieved at all by the local solver.

C. End Device Utility Function

After the definition of the communication model and FL model evaluation, we then give specific detail of the two-stage Stackelberg game for resource allocation [51]. Let c_n denote the number of required CPU cycles for device n to execute on a sample of data. The total number of CPU cycles required for end device n to run one local iteration is $c_n Q_n$, supposing all training data samples $\{x_i, y_i\}_{i \in Q_n}$ have the same size (i.e., pixel of the picture). Thus, the energy consumption of device n for one local iteration of model training satisfies [38]

$$E_n^k(\tau_n^k) = \alpha_n^k \tau_n^k c_n Q_n \delta_n^2 \quad (16)$$

where α_n denotes the effective capacitance coefficient of end device n 's computing chipset. In the k th round, all the end devices will first receive the price p_n^k from the ENs, the utility of end device n satisfies

$$u_n^k(\tau_n^k) = p_n^k(1 - \theta_n^k) - E_n^k(\tau_n^k) \quad (17)$$

where θ_n^k is the relative local model performance for device n in the k th round. According to (17), the first-order derivative of $u_n^k(\tau_n^k)$ w.r.t. τ_n^k is

$$\frac{\partial u_n^k}{\partial \tau_n^k} = \frac{p_n^k \psi(\phi_n Q_n)^v}{\tau^{(1-|\mathcal{A}|v)}} e^{-\psi(\phi_n Q_n \tau_n^{k|\mathcal{A}|})^v} - \alpha_n^k c_n Q_n \delta_n^2. \quad (18)$$

By setting $[(\partial u_n^k)/(\partial \tau_n^k)] = 0$, we obtain

$$h_n^k(\tau_n^{k*}) = p_n^k \quad (19)$$

where

$$h_n^k(\tau_n^{k*}) = \frac{\alpha_n^k c_n Q_n \delta_n^2}{\psi(\phi_n Q_n)^v} B_n^{k(1-|\mathcal{A}|v)} e^{\psi(\phi_n Q_n)^v \tau_n^{k(|\mathcal{A}|v)}}. \quad (20)$$

So the optimal τ_n^k can be obtained by

$$\tau_n^{k*} = h_n^{k-1}(p_n^k). \quad (21)$$

Theorem 1: The corresponding optimal τ_n^{k*} for p_n^k is unique.

Proof: Based on the first-order derivative of u_n^k , we can derive the second-order derivative of $u_n^k(\tau_n^k)$ w.r.t. τ_n^k as

$$\begin{aligned} \frac{\partial^2 u_n^k}{\partial (\tau_n^k)^2} &= p_n^k \left[(|\mathcal{A}|v - 1) \psi(\phi_n Q_n)^v \tau_n^{k(|\mathcal{A}|v-2)} \right. \\ &\quad \left. - \left(\psi(\phi_n Q_n)^v \tau_n^{k(|\mathcal{A}|v-1)} \right)^2 \right] e^{-\psi(\phi_n Q_n \tau_n^{k|\mathcal{A}|})^v} \end{aligned} \quad (22)$$

where $(|\mathcal{A}|v - 1) < 0$ is always satisfied, so $[(\partial^2 u_n^k)/(\partial (\tau_n^k)^2)] < 0$. ■

D. Edge Node Optimization Problem

The upload time for the model parameter will be $t_{n,\text{com}}^k = [(|\mathbf{w}|)/(C_n^k)]$, where $|\mathbf{w}|$ is the size of model parameters and C_n^k are the cooperative model transmission rate of device n . The time of computation $t_{n,\text{cmp}}^k$ for local training depends on the number of data batches trained by end device n , which can be calculated as $t_{n,\text{cmp}}^k = [(\tau_n^k c_n Q_n)/(\delta_n)]$.

Then, the total training time of one global iteration in the k th round of FL is defined as

$$T_{\text{global}}^k = \max_n \{t_{n,\text{com}}^k + t_{n,\text{cmp}}^k\}. \quad (23)$$

Thus, the training cost of the PS consisting of expected time cost and payment cost can be calculated as

$$U^k(p_1^k, \dots, p_n^k) = \lambda K(\Theta^k) T_{\text{global}}^k + \sum_{n=1}^N p_n^k (1 - \theta_n^k) \quad (24)$$

where λ is the time coefficient, Θ^k is the union of $[\theta_1^k, \dots, \theta_n^k]$, and $K(\Theta^k)$ is the bound improvement ratio in the k th round.

Theorem 2: The bound improvement ratio that means how many global iterations decrease according to more local iterations, can be satisfied as

$$K(\Theta^k) = \frac{\min_n \{1 - \theta_n(1)\}}{\min_n \{1 - \theta_n^k\}}. \quad (25)$$

Proof: To attain an ε -accuracy objective, there will always be an order of $\log(1/\varepsilon)$ global iterations needed [52]. For strongly convex objective $F(\mathbf{w})$, the general upper bound on global iterations is satisfied as

$$\begin{aligned} \mathcal{I}(\varepsilon, \Theta(\tau^k)) &= \max_n \frac{\mathcal{O}(\log(1/\varepsilon))}{1 - \theta_n(\tau_n^k)} \\ &= \frac{\mathcal{O}(\log(1/\varepsilon))}{\min_n \{1 - \theta_n(\tau_n^k)\}} \end{aligned} \quad (26)$$

where $\theta_n \in (0, 1]$ is the quality of local solution of device n . So, the bound improvement ratio can be calculated as

$$K(\Theta^k) = \frac{\mathcal{I}(\varepsilon, \Theta(\tau^k))}{\mathcal{I}(\varepsilon, \Theta(\mathbf{1}))} = \frac{\min_n \{1 - \theta_n(1)\}}{\min_n \{1 - \theta_n(\tau_n^k)\}}. \quad (27)$$

This bound improvement ratio is a powerful metric for a number of reasons. First, it allows devices to continue using their fine-tuned solvers, which can perform on a single device, instead of having to run completely new algorithms from scratch. Second, the actual performance evaluated by this metric is independent from the choice of optimization algorithm, making it much easier to optimize the overall performance. Third, based on this metric, the algorithm on one node network does not have to be optimal on another one, during the local update phase of FL. ■

Based on (24), the objective function of U^k is

$$\begin{aligned} U^k(\tau_1^{k*}, \dots, \tau_n^{k*}) &= \lambda \frac{\min_n \{1 - \theta_n(1)\}}{\min_n \{1 - \theta_n(\tau_n^{k*})\}} \\ &\quad \max_n \left\{ \frac{|\mathbf{w}|}{C_n^{k,d}} + \frac{|\mathbf{w}|}{C_n^{k,d}} + \frac{\tau_n^{k*} c_n Q_n}{\delta_n} \right\} \\ &\quad + \sum_{n=1}^N h(\tau_n^{k*}) (1 - \theta_n(\tau_n^{k*})). \end{aligned} \quad (28)$$

Theorem 3: The optimization problem $U^k(\tau^{k*})$ has a unique solution.

Proof: Take (28) into two parts. We will prove the convexity of (28). For the first part,

$$\max_n \left\{ \frac{\tau_n^{k*} c_n Q_n}{\delta_n} + \frac{|\mathbf{w}|}{C_n^*} \right\}$$

is linear increasing. From (14), we can easily derive that $\theta_n(\cdot)$ is a strictly convex function, which $1 - \theta_n(\cdot)$ is a strictly concave function. Hence, the reciprocal function is convex and nonincrease, so the first part is a strictly convex function. For the second part $g(\tau_n) = \sum_{n=1}^N h(\tau_n^{k*}) (1 - \theta_n(\tau_n^{k*}))$

$$\begin{aligned} g(\tau_n) &= h(\tau_n^{k*}) (1 - \theta_n(\tau_n^{k*})) \\ &= h(\tau_n^{k*}) - h(\tau_n^{k*}) \theta_n(\tau_n^{k*}) \\ &= h(\tau_n^{k*}) - \frac{\alpha_n^k c_n Q_n \delta_n^2}{\psi(\phi_n Q_n)^v} \tau_n^{k*(1-|\mathcal{A}|v)}. \end{aligned} \quad (29)$$

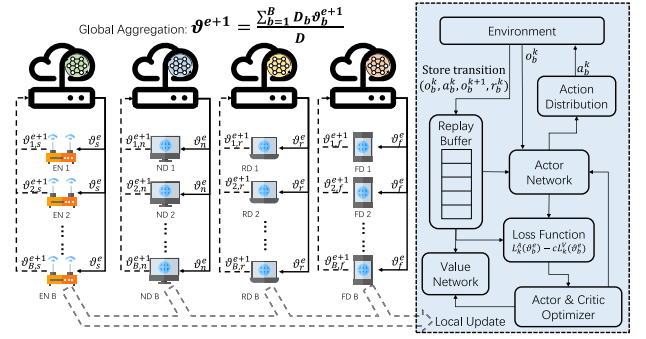


Fig. 2. Workflow of the MA-FRL algorithm consisting of local policy iteration and global aggregation.

From (21), we can easily derive that $h(\tau_n^{k*})$ is a convex function. For the second part of (29), because $0 < 1 - |\mathcal{A}|v < 1$, so $-\tau_n^{k*(1-|\mathcal{A}|v)}$ is also convex. Finally, the nonnegative finite weighted summation of $g(\tau_n)$ is also convex. ■

V. MULTIAGENT FEDERATED REINFORCEMENT LEARNING ALGORITHM

In this section, we describe how to transform the aforementioned problem into a learning task. To be specific, we model FL with multiple devices as a POMDP and design a multiagent federated reinforcement learning algorithm called MA-FRL to explore the optimization solution for the above secure incentivizing problem. By capturing the assumption of stationarity in this multiagent system, the proposed algorithm can dwindle variance in policy evaluation and then accelerate the convergence speed. The workflow of the proposed MA-FRL algorithm is shown in Fig. 2.

A. Federated Learning Model Design

1) *State Space:* We define the state space at the current decision round $k(k = 1, 2, \dots, K)$ as a union of the wireless channel conditions $G_b(t) = \{g_{b,n}^k, g_{b,r}^k, g_{b,f}^k\}$, the price strategies $P_b^k = \{p_{b,n}^k, p_{b,r}^k, p_{b,f}^k\}$, the local model performance $\Theta_b^k = \{\theta_{b,n}^k, \theta_{b,r}^k, \theta_{b,f}^k\}$, the number of local iterations $\tau_b^k = \{\tau_{b,n}^k, \tau_{b,r}^k, \tau_{b,f}^k\}$, the data quantity $Q_b^k = \{Q_{b,n}^k, Q_{b,r}^k, Q_{b,f}^k\}$, the computation of each data sample $C_b^k = \{c_{b,n}^k, c_{b,r}^k, c_{b,f}^k\}$, the data quality $\Phi_b^k = \{\phi_{b,n}^k, \phi_{b,r}^k, \phi_{b,f}^k\}$, and the computation capacity $\Delta_b^k = \{\delta_{b,n}^k, \delta_{b,r}^k, \delta_{b,f}^k\}$, which are denoted as

$$S_b^k := \{G_b^k, P_b^k, \Theta_b^k, \tau_b^k, Q_b^k, C_b^k, \Phi_b^k, \Delta_b^k\}. \quad (30)$$

2) *Partially Observable Policy:* To tackle the nonstationary problem in multiagent systems in FL, where agents may not share their observations or experiences with other agents due to privacy concerns. We formulate the POMDP for FL, so agents can only make decisions according to their local observations of the environment. We define the observation space of EN s in resource block b at the current decision round k as a union of the past L rounds local model performance of end devices and the past L rounds price profiles of EN s , which is denoted as $o_{b,s}^k := \{P_b^{k-L}, \Theta_b^{k-L}, \dots, P_b^{k-1}, \Theta_b^{k-1}\}$.

The EN's policy $\pi(a_{b,s}^k | o_{b,s}^k) \rightarrow [0, \infty)$ outputs the current pricing strategy $a_{b,s}^k := \{P_b^k\}$ w.r.t. current observation.

The observation of ND n consisting of the private information at current decision round k and the channel quality for the past L decision rounds, can be represented as $o_{b,n}^k := \{p_{b,n}^k, \tau_{b,n}^k, \phi_{b,n}^k, Q_{b,n}^k, c_{b,n}^k, \delta_{b,n}^k, g_{b,n}^{k-L}, \dots, g_{b,n}^{k-1}\}$. The ND n allocates jamming signal to protect model transmission during the upload phase that its policy can be represented as $\pi(a_{b,n}^k | o_{b,n}^k) \rightarrow [0, 1]$, where its action is $a_{b,n}^k := \{\rho_{b,n}^{k,d}, \Omega_{b,n}^{k,d}, \rho_{b,n}^{k,u}\}$. The policy of RD r can be represented as $\pi(a_{b,r}^k | o_{b,r}^k) \rightarrow [0, 1]$, where the observation is defined as $o_{b,r}^k := \{p_{b,r}^k, \tau_{b,r}^k, \phi_{b,r}^k, Q_{b,r}^k, c_{b,r}^k, \delta_{b,r}^k, g_{b,r}^{k-L}, \dots, g_{b,r}^{k-1}\}$ and the action is $a_{b,r}^k := \{\rho_{b,r}^{k,d}, \rho_{b,r}^{k,u}\}$. For FD f with policy $\pi(a_{b,f}^k | o_{b,f}^k) \rightarrow [0, 1]$, its observation is $o_{b,f}^k := \{p_{b,f}^k, \tau_{b,f}^k, \phi_{b,f}^k, Q_{b,f}^k, c_{b,f}^k, \delta_{b,f}^k, g_{b,f}^{k-L}, \dots, g_{b,f}^{k-1}\}$ and its action is $a_{b,f}^k := \{\rho_{b,f}^{k,d}, \Omega_{b,f}^{k,u}, \rho_{b,f}^{k,u}\}$.

3) *Reward*: After the state transition, ENs and end devices would gain rewards w.r.t. current state S_b^k and corresponding actions. The reward function of ENs can be defined as

$$R(S_b^k, a_{b,s}^k) = -U_b^k \quad (31)$$

where U_b^k is the expected system cost of resource block b computed in (24). Besides, the reward of end device n w.r.t. the utility computed in (17) can be defined as

$$R(S_b^k, a_n^k) = u_n^k - \alpha_p l_p \quad (32)$$

where l_p is the privacy leakage probability in (13) and α_p is the leakage penalty coefficient.

4) *Value Function*: Given policy π , the value function $V_\pi(S)$ of the state S , the expected return when starting in S and following π thereafter, can be formally by

$$V_\pi(S) := \mathbb{E}_\pi \left[\sum_{k=0}^K \gamma^k R(S^k, a^k) | S^0 = S \right] \quad (33)$$

where $\mathbb{E}_\pi(\cdot)$ denotes the expected value of a random variable given that the agent follows policy π and $\gamma \in [0, 1]$ is the reward discount factor used to reduce the weights as the time step increases.

B. Local Policy Iteration

The computing of value function can help each agent to look for a better policy. When there is a determined value function V_π for an arbitrary policy π , we can improve the policy through policy iteration. The key idea behind the policy gradient is to increase the probability of actions that lead to higher returns and reduce the probability of actions that lead to lower returns until we reach the optimal policy. We use the neural network parameter ϑ_b to parameterize the policy and value function of the agents in resource block b . At each training iteration, we randomly select samples from the replay buffer to update the network parameters. We use generalized advantage estimation [53] to compute variance-reduced advantage-function estimator $A(S, a)$ that makes use of a learning state-value function $V(S)$. We then update the

policy and value function by maximizing the objective via stochastic gradient ascent as follows:

$$\vartheta_b^{e+1} = \arg \max_{\vartheta_b} \frac{1}{D_b} \sum \mathbb{E}_k [L_k^A(\vartheta_b^e) - cL_k^{VF}(\vartheta_b^e)] \quad (34)$$

$$L_k^A(\vartheta_b^e) = \min \left[r_b^e(\vartheta_b^e) A(S_b^k, a_b^k), g(\epsilon, A(S_b^k, a_b^k)) \right] \quad (35)$$

$$L_k^{VF}(\vartheta_b^e) = (V(S_b^k) - V_{\text{targ}}^k)^2 \quad (36)$$

where

$$r_b^e(\vartheta_b^e) = \frac{\pi_{\vartheta_b^e}(a_b^k | o_b^k)}{\pi_{\vartheta_b^e}(a_b^k | o_b^k)} \quad (37)$$

$$A(S_b^k, a_b^k) = -V(S_b^k) + \sum_{l=k}^{K-1} \gamma^{l-k} R(S_b^l, a_b^l) + \gamma^{K-k} V(S_b^K) \quad (38)$$

$$g(\epsilon, A) = \begin{cases} (1 + \epsilon)A & A \geq 0 \\ (1 - \epsilon)A & A < 0 \end{cases} \quad (39)$$

and V_{targ}^k is the total discount reward from k to the end of episode, $\vartheta_b^{e'}$ represents the parameter of policy for sampling, r_b^k is the importance ratio, and c is the loss coefficient of the value function. Because computing and storage capabilities of the end devices within systems are heterogeneous, the number of iterations D_b for resource block $b \in \mathcal{B}$ can be different.

C. Global Policy Aggregation

As mentioned above, the optimal strategy can be dynamically and efficiently found by MARL. However, there are still some challenges for the direct deployment of MARL in a privacy-preserving system.

- 1) *Unshared Experience*: Agents' experiences may contain private information that cannot be shared with others, especially in commercial and industrial scenarios. However, there are limited samples that can be used to train and evaluate the local policy. Almost all the real-world systems are fragile, or expensive enough that the data they produce is costly. As the exploratory actions do not come for free, agents cannot have a separate exploration policy during training.
- 2) *Unshared Observation*: For the protection of their privacy, agents may not share their partial observation with other agents in the environment, that is, each agent can only make decisions based on their partial observation. Training each agent from scratch would cause the non-stationary and take a long time or even impossible to converge.

Motivated by the aforementioned reasons, we propose a global policy. The aggregation process via FedAvg [29] of the RL policy can be expressed as

$$\vartheta^{e+1} = \frac{\sum_{b=1}^B D_b \vartheta_b^{e+1}}{D} \quad (40)$$

where $D = \sum_{b=1}^B D_b$ is the number of total experience. And the updated global policy ϑ^{e+1} is used to the next-episode training.

Denote ϑ^* as the optimal solution. We have the following assumptions as in [54].

Assumption 1: For all $L_b(\vartheta_b) = L_k^A(\vartheta_b) - cL_k^{VF}(\vartheta_b)$:

- 1) $L_b(\vartheta_b)$ is convex;
- 2) $L_b(\vartheta_b)$ is β -smooth, i.e., $L_b(\vartheta'_b) \leq L_b(\vartheta_b) + \nabla L_b(\vartheta_b) \cdot (\vartheta'_b - \vartheta_b) + (\beta/2)\|\vartheta_b - \vartheta'_b\|^2$, for $\forall \vartheta_b$ and ϑ'_b .

The feasibility of the linear regression and the update rule of FL is guaranteed by Assumption 1. We then have the following lemma.

Lemma 1: $L(\vartheta) = \sum_{b=1}^B D_b L_b(\vartheta_b)/D$ is μ -strongly convex and β -smooth.

Proof: Please see Appendix A. ■

Theorem 4: Considering that $L(\vartheta)$ is β -smooth and μ -strongly, let $\eta = 1/L$ and $\vartheta^* = \arg \min_{\vartheta} L(\vartheta)$, we have

$$\|\vartheta^e - \vartheta^*\| \leq \left(1 - \frac{\mu}{\beta}\right)^e \|\vartheta^1 - \vartheta^*\| \quad (41)$$

so the gradient dispersion can be derived as $O(\bar{\vartheta}) = (\beta/\mu) \log(\|\vartheta^1 - \vartheta^*\|/\bar{\vartheta})$, which is used to illustrate how the parameters ϑ^t are distributed in each participant.

Proof: Please see Appendix B. ■

The convergence in expectations can be formulated as

$$[L(\vartheta^e) - L(\vartheta^*)] \leq \bar{\vartheta}^e [\Delta^e(L(\vartheta^*))]. \quad (42)$$

And $L(\vartheta)$ is proven to be bounded where $\Delta^e(L(\vartheta^*)) = L(\vartheta^1) - L(\vartheta^*)$. Our proposed MA-FRL algorithm can find the optimal secure resource allocation strategy without sharing information and experience, with the experience implied in the aggregated global model. As for the complexity of the MA-FRL algorithm, each device maintains its own policy and performs decisions independently during the decision-making phase. In this way, the input dimension and output dimension are constant determined by its observation space and action space. The number of hidden neurons in each hidden layer is set to U , and the number of hidden layers is set to L . Thereby the complexity analysis is $O(UL)$ in each decision slot. Algorithm 1 illustrates the overall workflow of the proposed MA-FRL algorithm for secure incentive mechanism in FL.

VI. NUMERICAL RESULTS

In this section, we evaluate the performance of the proposed multiagent learning-based incentive mechanism via extensive both real-world and simulation experiments.

A. Simulation Settings

In the experiment, the system consists of a PS, five ENs, and 15 end devices. There are five resource blocks, and each resource block contains an EN, an ND, an RD, and an FD. The system is set in a full participation fashion, which means all the devices, whether they have more or fewer data, have to participate in model training. For the computation cost for end devices, α_n is set to 0.03, the c_n is set to 0.05, and the δ is set to 2. The global time coefficient λ is set to 0.01 to scale the time cost. As the adaptive downlink and uplink cooperative jamming schemes in cooperative NOMA proposed in [46], the parameters of the wireless communication model for collaborative federation learning are set as follows. The SNR $\rho = 30$ for one resource block, the channel bandwidth is set

Algorithm 1: Proposed MA-FRL Algorithm for Secure Incentive Mechanism

```

1 Initialize  $\mathbf{w}, \vartheta_s, \vartheta_n, \vartheta_r, \vartheta_f, \gamma, \eta, \epsilon$ 
2 for Episode  $e \in 1, \dots, E$  do
3   for  $b \in \mathcal{B}$  do
4     Reset  $\mathbf{w}_b^0, S_b^0$ .
5     Reset replay buffer  $\mathcal{D}_{b,s}, \mathcal{D}_{b,n}, \mathcal{D}_{b,r}, \mathcal{D}_{b,f}$ .
6      $\vartheta_{b,s}^e = \vartheta_s^e, \vartheta_{b,n}^e = \vartheta_n^e, \vartheta_{b,r}^e = \vartheta_r^e, \vartheta_{b,f}^e = \vartheta_f^e$ .
7   end
8   for Round  $k \in 1, \dots, K$  do
9     In each resource block  $b$ , each EN observes  $S_b^k$  and
      updates its observation to  $o_{b,s}^k$ .
10    Input  $o_{b,s}^k$  into each EN's actor policy  $\vartheta_{b,s}^e$  and
      determines the current pricing strategy  $P_b^k$  for end
      devices.
11    End devices determine their number of local iterations
       $\tau_b^k$  by (21).
12    End devices observer  $S_b^k$  and update their observations
      to  $o_{b,n}^k, o_{b,r}^k, o_{b,f}^k$ .
13    Input  $o_{b,n}^k, o_{b,r}^k, o_{b,f}^k$  to actor network  $\vartheta_{b,n}^e, \vartheta_{b,r}^e, \vartheta_{b,f}^e$ 
      and determine transmit power  $\rho_{b,n}^k, \rho_{b,r}^k, \rho_{b,f}^k$  and
      jamming coefficient  $\Omega_{b,d}^k, \Omega_{b,u}^k$ .
14    End devices download  $\mathbf{w}^k$  from the parameter server
      while far devices allocate jamming signals.
15    End devices perform  $\tau_b^k$  round local model update.
16    End devices upload  $\mathbf{w}_n^{k+1}$  to the parameter server
      while near devices allocate jamming signals.
17    PS aggregated global model for the end devices with
       $\mathbf{w}^{k+1} = \frac{\sum_{n=1}^N Q_n \mathbf{w}_n^{k+1}}{Q}$ .
18    Updates the  $S_b^k$  into  $S_b^{k+1}$  and calculates rewards
       $R_{b,s}^k, R_{b,n}^k, R_{b,r}^k, R_{b,f}^k$ .
19    Store transitions  $(o_{b,s}^k, a_{b,s}^k, R_{b,s}^k, o_{b,s}^{k+1})$  in  $\mathcal{D}_{b,s}$ ,
       $(o_{b,n}^k, a_{b,n}^k, R_{b,n}^k, o_{b,n}^{k+1})$  in  $\mathcal{D}_{b,n}$ ,  $(o_{b,r}^k, a_{b,r}^k, R_{b,r}^k, o_{b,r}^{k+1})$ 
      in  $\mathcal{D}_{b,r}$ ,  $(o_{b,f}^k, a_{b,f}^k, R_{b,f}^k, o_{b,f}^{k+1})$  in  $\mathcal{D}_{b,f}$ .
20   end
21   for  $b \in \mathcal{B}$  do
22     Sample  $D_b$  transitions from  $\mathcal{D}_{b,s}, \mathcal{D}_{b,n}, \mathcal{D}_{b,r}, \mathcal{D}_{b,f}$ .
23     Update  $\vartheta_{b,s}^e, \vartheta_{b,n}^e, \vartheta_{b,r}^e, \vartheta_{b,f}^e$  through (34) and upload
      to PS.
24   end
25   PS aggregates global models  $\vartheta_s^{e+1}, \vartheta_n^{e+1}, \vartheta_r^{e+1}, \vartheta_f^{e+1}$ 
      according to (40).
26 end

```

to $W = 20$, and the path-loss factor is set to 2. In simulation experiments, the default value for data quality is $\phi = 0.5$, and the default value for data quantity is $Q = 512$. We conduct our experiments using PyTorch 1.8 on Windows 10. To represent the policy, we used a fully connected multilayer processor with two hidden layers of 64 units and output headers. For servers actor network, there are three headers to determine its pricing strategy. When it comes to device actor network, there are three headers for near and FDs and two headers for RDs to determine their communication strategies. As for the proposed MA-FRL-based approach, the local learning rate is set to 0.0003, the discount factor is $\gamma = 0.99$, the clip ratio is $\epsilon = 0.2$, and the value coefficient is $c = 0.5$. The number of training episodes is set to $E = 100$, and the length of each

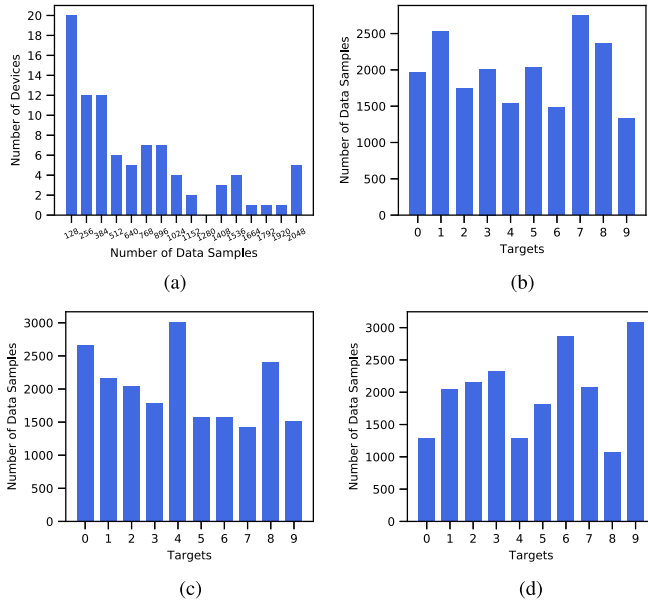


Fig. 3. FMNIST data set description. (a) Unbalanced data distribution. (b) Data distributed in NDs. (c) Data distributed in RDs. (d) Data distributed in FDs.

episode is $K = 16$ and the number of past experiences is $L = 4$. When performing the local update, each device can choose a different $D_b \in \{16, 32\}$ of minibatch size according to its computation resource. Finally, the Adam optimizer is used to optimize the neural networks. Our code is available at: <https://github.com/Lucky454/MAFRL>.

Moreover, to demonstrate the effectiveness of the MA-FRL-based approach, we perform it on the real-world federated data set FMNIST. The data set contains 60 000 training examples and 10 000 test examples, which can be used to perform a digit classification task. Fig. 3 shows the statistics of the data distribution in the system. First, the data examples distribution among different devices is unbalancing. As shown in Fig. 3(a), the majority of devices have minority data examples, while a small number of devices have lots of data examples. However, there are many devices with a small number of data examples in the real world. Second, the data examples used by the devices in each local model training are nonindependent identically distributed. As shown in Fig. 3(b)–(d), there are some differences in the distribution and quality of data collected by devices in different locations, due to different user preferences.

In the experiments, the metrics used to evaluate the performance of the proposed approach can be summarized as follows.

1) *Training Cost*: This is an economic metric to evaluate the global performance of FL. As defined in (24), the training cost in the k th round consists of expected time cost and payment cost. To minimize the training cost of the server is one of the objects of the proposed approach.

2) *Objective Divergence*: This is an economic metric used to characterize the personality in the local model of end devices, which can be calculated as

$$\mathcal{D} = \sqrt[4]{\frac{1}{N} \sum_{n=1}^N (u_n^k - \bar{u}^k)^2} \quad (43)$$

where $\bar{u}^k$ is the mean utility of the end devices in the k th round.

3) *Leak Probability*: As computed in (13), this is a privacy metric that the probability of privacy information is eavesdropped by untrusted RDs during model updates transmitting.

B. Convergence Analysis

In this work, we propose an MA-FRL algorithm that allows multiple agents to learn global policies collaboratively while without privacy experience sharing. In this way, the proposed algorithm can accelerate global learning processes and overcomes nonstationary problems in the proposed multiagent incentive mechanism faster. To demonstrate this, we compare the convergence properties of the MA-FRL algorithm with the multiagent proximal policy optimization algorithm (MAPPO) [18] and asynchronous advantage actor-critic (A3C) [55]. MAPPO is a state-of-the-art MARL algorithm that utilizes a KL penalty coefficient to adjust stepsizes of parameter update adaptively. On the other hand, A3C is a state-of-the-art DRL algorithm that allows multiple environments to collect experience asynchronously and trains policies on this experience centralized. Moreover, to show how different numbers of resource blocks affect the learning process of the proposed algorithm, we also compare the performance when the number of resource blocks equals 3 and 5, which is denoted as MA-FRL-3 and MAFRL-5. In the meanwhile, the A3C algorithm collecting experience from five different environments is denoted as A3C-5. Fig. 4 shows the time cost, objective divergence, and leak probability under previous settings and real-world data sets.

As shown in Fig. 4(a), the MA-FRL-5 takes about 25 episodes for convergence to a stable training cost performance while MA-FRL-3 takes about 35 episodes. A3C-5 is the fastest algorithm for convergence, but its training cost is a little bit lower than other algorithms. As we can see, the performance of MAPPO is the slowest algorithm, which takes about 60 episodes for convergence. That is because there is limited experience generated from the environment for MAPPO to evaluate and improve its police. However, the other three algorithms can utilize more experience generated from multiple environments to accelerate their learning process. From Fig. 4(b), we can observe that the objective divergence converges during the training phase. A3C-5 is the fastest algorithm to converge in this context. The convergence speed of MAPPO and MA-FRL-3 is similar. These three algorithms all converge to a pretty high objective divergence at about 0.65 even MA-FRL-3 discovers some low-quality experience during training. MA-FRL-5 converges to the lowest objective divergence at about 0.4. As shown in Fig. 4(c), the speeds of these four algorithms to converge to low leak probability are similar. Aside from the converging rate of these four algorithms, the station of performance is also important to evaluate these algorithms. As we can observe from Fig. 4, in the beginning phase, i.e., the exploration phase, the performance of MAPPO is always more unstable than the other three algorithms. This phenomenon can be explained by FL in a multiagent system environment. During the exploration phase, agents not only receive the environment but are also affected

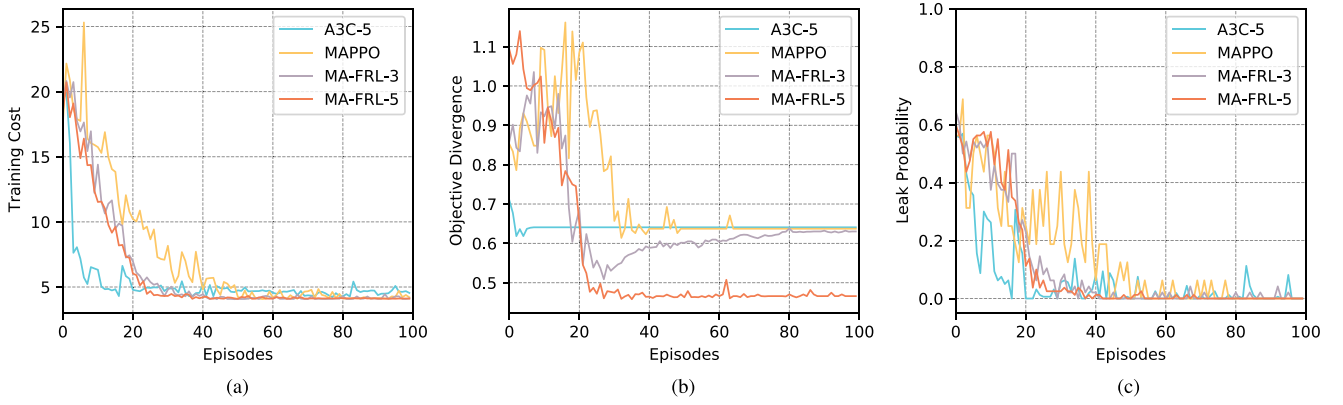


Fig. 4. Convergence analysis of the MA-FRL-based approach. (a) Training cost versus episodes. (b) Objective divergence versus episodes. (c) Leak probability versus episodes.

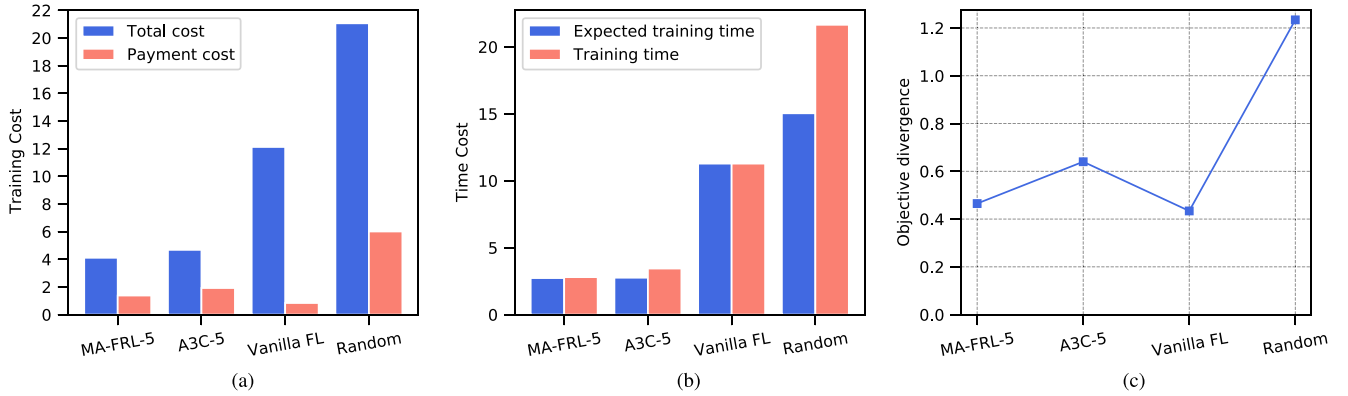


Fig. 5. Performance comparison under real-world data sets. (a) Training cost in each round. (b) Time cost in each round. (c) Objective divergence in each round.

by other agents' decisions. As a result, it is difficult for them to have a good understanding of the outer environment. But as training goes, they can finally learn some rules and end up reaching equilibrium.

Comparing the MA-FRL-5 and MA-FRL-3 in Fig. 4, MA-FRL-5 converges faster than MA-FRL-3 because it can drive more agents to interact with the environment and gains more experience. The more experience for algorithms used in training, the faster their convergence will be. In addition, in Fig. 5(b), the convergence result of the MA-FRL-5 algorithm is better than that of MA-FRL-3, and the important reason for that is MA-FRL-5 can explore a more divergent strategy set and thus converges to a lower objective divergence.

C. Performance Evaluation Under Real-World Data Sets

In this section, we compare the proposed MA-FRL-based approach with other baselines (A3C-5, Vanilla FL [10], and Random) algorithm under previous systems setting. In the Vanilla FL approach, end devices perform a single local update in each round and then upload to the server in threshold secure transmission rate. The Random approach uses the same transmission rate, but the number of local updates is chosen randomly. Fig. 5(a) shows the training costs and payment costs of these algorithms in each round. We can observe that the Random approach costs the highest payment and the Vanilla

FL approach costs the lowest payment, but their total costs are still high. We can observe from Fig. 5(b) that the MA-FRL-based approach pays two more times payments than Vanilla FL, but its training time is only 20% of that of Vanilla FL. As shown in Fig. 4(c), the devices' objective divergences of the proposed approach are similar to Vanilla FL but have more outstanding global performance.

D. Performance Evaluation Under Various Environment Settings

Fig. 6 shows that the training cost of each algorithm fluctuates with the increase of data quantity. The larger data quantity will increase training costs under the same computation capability of devices. Compared with MA-FRL-5, the Vanilla FL approach increases more severely. It is interesting to notice that when data quantity equals 500, the training cost of A3C-5 is much higher than that of MA-FRL-5. But when data quantity becomes larger, this situation is flipped. When data quantity equals 900, the training cost of A3C-5 is a little lower than that of MA-FRL-5. Under various data quantities, the training cost of Vanilla FL is always five times higher than that of MA-FRL-5 and A3C-5.

Fig. 7 shows that the training cost increases when the data quality increases. Due to the performance of the FL model largely depending on its data quality, if data quality is high,

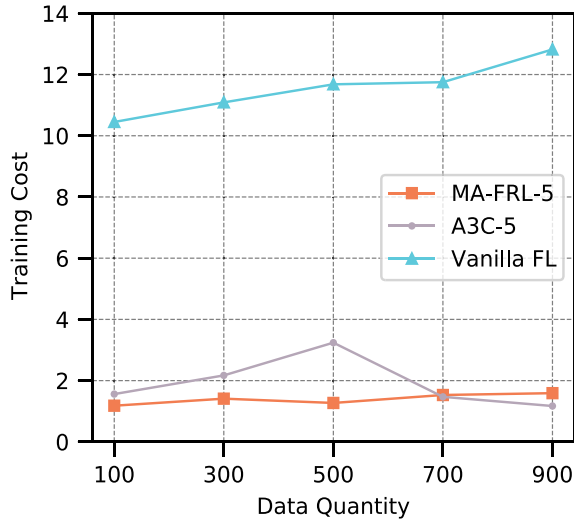


Fig. 6. Training cost versus data quantity.

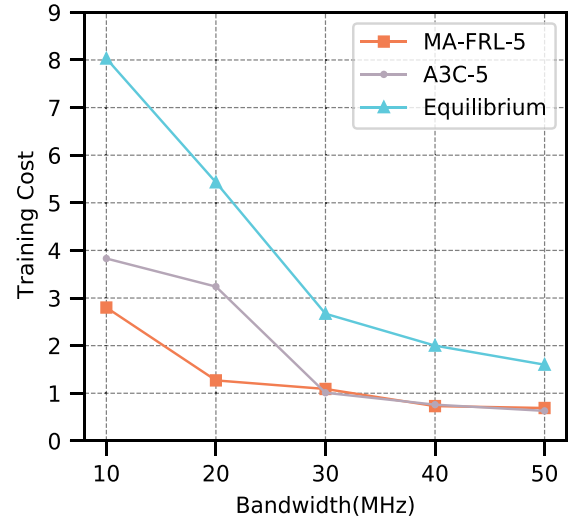


Fig. 8. Training cost versus system bandwidth.

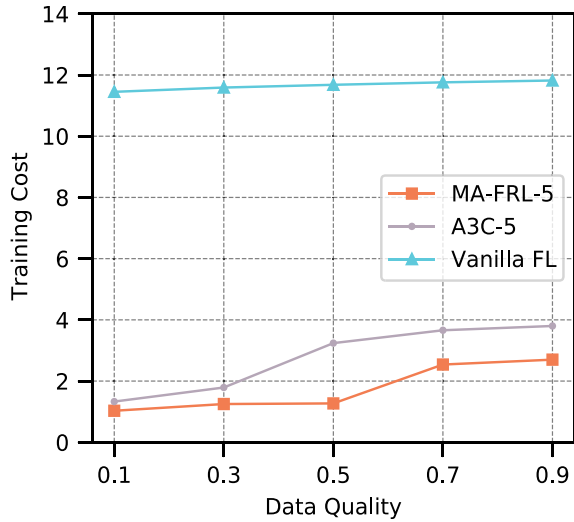
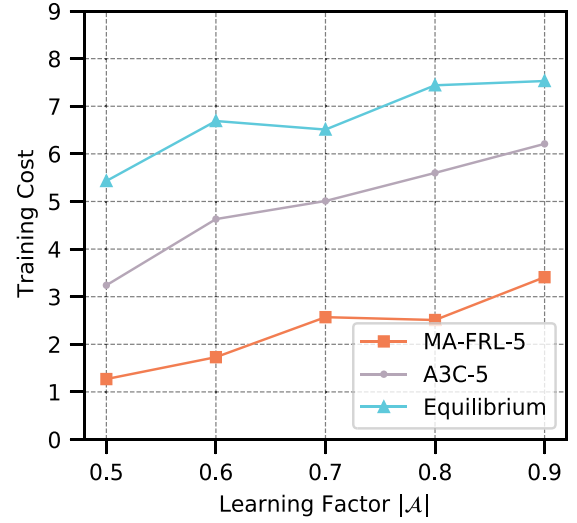


Fig. 7. Training cost versus data quality.

Fig. 9. Training cost versus learning factor $|A|$.

devices can train better models and receive higher payment from the server. When the data quality is low, the performances of MA-FRL-5 and A3C-5 are similar. As data quality increases, the performance between these two algorithms also increases, and the peak occurs when data quality equals 0.5. Similar to different data quality settings, in the data quality context, the training cost of the Vanilla FL approach is always higher than that of MA-FRL-5 and A3C-5. But when data quality is high, such as 0.7 and 0.9, the performance gap between these three approaches is smaller.

As illustrated in Fig. 8, the simulations run under scenarios where the channel bandwidth in the system ranges from 10 to 50 MHz. In these simulations, the Equilibrium approach is used to compare with the proposed approach. The Equilibrium approach is the Stackelberg equilibrium of this multiagent game under threshold secure transmission rate. As the channel bandwidth allocated by the system increase, the relative benefit of the proposed approach decreases. The reason behind it is similar to the benefit of distributed machine learning, if there is enough system bandwidth, we can utilize

this resource to perform a large number of global updates and just do a few local updates to save computation.

We then present the performance under different learning factors from different learning algorithms in Fig. 9. The learning factor of an algorithm means how fast this algorithm fits a specific goal. As we can observe, as the learning factor increases, the training cost of the server also increases. That is because devices with higher learning factors can obtain a better model in local training, which can be sold at a higher price in trading with the server. However, compared with the other two algorithms, the training cost of the MA-FRL-5 approach is always lowest under various learning factors.

VII. CONCLUSION

In this article, we have studied the MA-FRL for secure incentive mechanism in ICPS with heterogeneous devices. We first propose a secure collaborative communication model, where FDs in the download phase and NDs in the upload phase are recruited to emit jamming signals adaptively to prevent information leakage at untrusted relays. Then, we formulate the communication, computation, data resource

allocation problem as a Stackelberg game in FL with heterogeneous devices. To address the dynamics and nonstationarity of this environment, we model this multiagent system as a POMDP. In particular, we propose the MA-FRL algorithm that learns to solve the nonstationary problem caused by frequent interaction among devices in FL without privacy information sharing. Furthermore, we prove that MA-FRL is μ -strongly convex and β -smooth and derive its convergence rate in expectation. Numerical results have indicated that the proposed MA-FRL-based approach matches the theoretical analysis and outperforms baseline approaches. For future work, a more efficient federated reinforcement learning algorithm that can take essential policy updates and discard gross ones can be considered to reduce communication overhead caused by policy updates and improve the rate of convergence. In the meanwhile, we also want to utilize the blockchain to evolve the current MA-FRL algorithm into a decentralized manner.

APPENDIX A PROOF OF LEMMA 1

Straightforwardly, from Assumption 1, according to the definition of convex and Theorem 3, $L(\vartheta)$ is the finite-sum structure of $L_b(\vartheta)$, and the triangle inequality.

APPENDIX B PROOF OF THEOREM 4

According to the μ -strongly convexity of $L(\vartheta)$, we have

$$\nabla L(\vartheta)(\vartheta - \vartheta^*) \geq L(\vartheta) - L(\vartheta^*) + \frac{\mu}{2} \|\vartheta - \vartheta^*\|^2. \quad (44)$$

Thus, we can obtain the following:

$$\begin{aligned} \|\vartheta^{e+1} - \vartheta^*\|^2 &= \|\vartheta^e - \eta \nabla L(\vartheta^e) - \vartheta^*\|^2 \\ &= \|\vartheta^e - \vartheta^*\|^2 - 2\eta \nabla L(\vartheta^e)(\vartheta^e - \vartheta^*) \\ &\quad + \eta^2 \|\nabla L(\vartheta^e)\|^2 \\ &\leq \|\vartheta^e - \vartheta^*\|^2 - 2\eta \\ &\quad \times \left(L(\vartheta) - L(\vartheta^*) + \frac{\mu}{2} \|\vartheta^e - \vartheta^*\|^2 \right) \\ &\quad + \eta^2 \|\nabla L(\vartheta^e)\|^2. \end{aligned} \quad (45)$$

By smoothing $L(\vartheta)$, we can obtain the gradient bound

$$\begin{aligned} L(\vartheta^*) &\leq L\left(\vartheta - \frac{1}{\beta} \nabla L(\vartheta)\right) \\ &\leq L(\vartheta) - \|\nabla L(\vartheta)\|^2 + \frac{1}{2\beta} \|\nabla L(\vartheta)\|^2 \\ &\leq L(\vartheta) - \frac{1}{2\beta}. \end{aligned} \quad (46)$$

By incorporating (45), (46) can be transformed as

$$\begin{aligned} \|\vartheta^{e+1} - \vartheta^*\| &= \|\vartheta^e - \eta \nabla L(\vartheta^e) - \vartheta^*\| \\ &\leq \|\vartheta^e - \vartheta^*\| - \eta \mu \|\vartheta^{e+1} - \vartheta^*\| + 2\eta(\eta\beta - 1) \\ &\quad \times (L(\vartheta) - L(\vartheta^*)) \\ &\leq \left(1 - \frac{\mu}{\beta}\right) \|\vartheta^e - \vartheta^*\| \leq \left(1 - \frac{\mu}{\beta}\right) \|\Delta^*(\vartheta)\| \end{aligned} \quad (47)$$

where η is set as the last step.

REFERENCES

- [1] S. Rathore and J. H. Park, "A blockchain-based deep learning approach for cyber security in next generation industrial cyber-physical systems," *IEEE Trans. Ind. Informat.*, vol. 17, no. 8, pp. 5522–5532, Aug. 2021.
- [2] C. Wang, G. Yang, G. Papanastasiou, H. Zhang, J. J. P. C. Rodrigues, and V. Albuquerque, "Industrial cyber-physical systems-based cloud IoT edge for federated heterogeneous distillation," *IEEE Trans. Ind. Informat.*, vol. 17, no. 8, pp. 5511–5521, Aug. 2021.
- [3] Y. Lu, X. Huang, Y. Dai, S. Maharjan, and Y. Zhang, "Federated learning for data privacy preservation in vehicular cyber-physical systems," *IEEE Netw.*, vol. 34, no. 3, pp. 50–56, May/Jun. 2020.
- [4] C. Zhang, X. Liu, X. Zheng, R. Li, and H. Liu, "FengHuoLun: A federated learning based edge computing platform for cyber-physical systems," in *Proc. IEEE Int. Conf. Pervasive Comput. Commun. Workshops (PerCom Workshops)*, Austin, TX, USA, 2020, pp. 1–4.
- [5] Y. Liu, J. J. Q. Yu, J. Kang, D. Niyato, and S. Zhang, "Privacy-preserving traffic flow prediction: A federated learning approach," *IEEE Internet Things J.*, vol. 7, no. 8, pp. 7751–7763, Aug. 2020.
- [6] H. Peng and X. S. Shen, "Deep reinforcement learning based resource management for multi-access edge computing in vehicular networks," *IEEE Trans. Netw. Sci. Eng.*, vol. 7, no. 4, pp. 2416–2428, Oct./Dec. 2020.
- [7] Z. Hong, W. Chen, H. Huang, S. Guo, and Z. Zheng, "Multi-hop cooperative computation offloading for industrial IoT-edge-cloud computing environments," *IEEE Trans. Parallel Distrib. Syst.*, vol. 30, no. 12, pp. 2759–2774, Dec. 2019.
- [8] M. Chen, H. V. Poor, W. Saad, and S. Cui, "Wireless communications for collaborative federated learning in the Internet of Things," 2020. [Online]. Available: arXiv:2006.02499.
- [9] I. A. Elgendy, W.-Z. Zhang, H. He, B. B. Gupta, and A. A. Abd El-Latif, "Joint computation offloading and task caching for multi-user and multi-task MEC systems: Reinforcement learning-based algorithms," *Wireless Netw.*, vol. 27, pp. 2023–2038, Feb. 2021.
- [10] Q. Yang, Y. Liu, T. Chen, and Y. Tong, "Federated machine learning: Concept and applications," *ACM Trans. Intell. Syst. Technol.*, vol. 10, no. 2, pp. 1–19, 2019.
- [11] W. Y. B. Lim *et al.*, "Towards federated learning in UAV-enabled Internet of Vehicles: A multi-dimensional contract-matching approach," 2020. [Online]. Available: arXiv:2004.03877.
- [12] J. Kang, Z. Xiong, D. Niyato, Y. Zou, Y. Zhang, and M. Guizani, "Reliable federated learning for mobile networks," *IEEE Wireless Commun.*, vol. 27, no. 2, pp. 72–80, Apr. 2020.
- [13] N. H. Tran, W. Bao, A. Zomaya, Minh N. H. Nguyen, and C. S. Hong, "Federated learning over wireless networks: Optimization model design and analysis," in *Proc. IEEE INFOCOM Conf. Comput. Commun.*, Paris, France, 2019, pp. 1387–1395.
- [14] S. Wang *et al.*, "When edge meets learning: Adaptive control for resource-constrained distributed machine learning," in *Proc. IEEE INFOCOM Conf. Comput. Commun.*, Honolulu, HI, USA, 2018, pp. 63–71.
- [15] J. Kang, Z. Xiong, D. Niyato, S. Xie, and J. Zhang, "Incentive mechanism for reliable federated learning: A joint optimization approach to combining reputation and contract theory," *IEEE Internet Things J.*, vol. 6, no. 6, pp. 10700–10714, Dec. 2019.
- [16] W. Y. B. Lim *et al.*, "Hierarchical incentive mechanism design for federated machine learning in mobile networks," *IEEE Internet Things J.*, vol. 7, no. 10, pp. 9575–9588, Oct. 2020.
- [17] W. Y. B. Lim *et al.*, "When information freshness meets service latency in federated learning: A task-aware incentive scheme for smart industries," *IEEE Trans. Ind. Informat.*, early access, Dec. 21, 2020, doi: 10.1109/TII.2020.3046028.
- [18] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017. [Online]. Available: arXiv:1707.06347.
- [19] X. Qiu, L. Liu, W. Chen, Z. Hong, and Z. Zheng, "Online deep reinforcement learning for computation offloading in blockchain-empowered mobile edge computing," *IEEE Trans. Veh. Technol.*, vol. 68, no. 8, pp. 8050–8062, Aug. 2019.
- [20] Y. He, N. Zhao, and H. Yin, "Integrated networking, caching, and computing for connected vehicles: A deep reinforcement learning approach," *IEEE Trans. Veh. Technol.*, vol. 67, no. 1, pp. 44–55, Jan. 2018.
- [21] Y. He *et al.*, "Deep-reinforcement-learning-based optimization for cache-enabled opportunistic interference alignment wireless networks," *IEEE Trans. Veh. Technol.*, vol. 66, no. 11, pp. 10433–10445, Nov. 2017.

- [22] A. Ferdowsi, U. Challita, W. Saad, and N. B. Mandayam, "Robust deep reinforcement learning for security and safety in autonomous vehicle systems," in *Proc. 21st Int. Conf. Intell. Transp. Syst. (ITSC)*, Maui, HI, USA, 2018, pp. 307–312.
- [23] H. Peng and X. Shen, "Multi-agent reinforcement learning based resource management in MEC- and UAV-assisted vehicular networks," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 1, pp. 131–141, Jan. 2021.
- [24] P. Hernandez-Leal, B. Kartal, and M. E. Taylor, "Is multiagent deep reinforcement learning the answer or the question? a brief survey," *Learning*, vol. 21, p. 22, Oct. 2018.
- [25] C. L. Stergiou, K. E. Psannis, and B. B. Gupta, "IoT-based big data secure management in the fog over a 6G wireless network," *IEEE Internet Things J.*, vol. 8, no. 7, pp. 5164–5171, Apr. 2021.
- [26] Z. A. Al-Sharif, M. I. Al-Saleh, L. M. Alawneh, Y. I. Jararweh, and B. Gupta, "Live forensics of software attacks on cyber-physical systems," *Future Gener. Comput. Syst.*, vol. 108, pp. 1217–1229, Jul. 2020.
- [27] W.-Z. Zhang *et al.*, "Secure and optimized load balancing for multitier IoT and edge-cloud computing systems," *IEEE Internet Things J.*, vol. 8, no. 10, pp. 8119–8132, May 2021.
- [28] H. H. Zhuo, W. Feng, Q. Xu, Q. Yang, and Y. Lin, "Federated reinforcement learning," 2019. [Online]. Available: arXiv:1901.08277.
- [29] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. Y. Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. Int. Conf. Artif. Intell. Stat.*, 2017, pp. 1273–1282.
- [30] W. Y. B. Lim *et al.*, "Federated learning in mobile edge networks: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 3, pp. 2031–2063, 3rd Quart., 2020.
- [31] Y. Liu, J. Peng, J. Kang, A. M. Iliyasu, D. Niyato, and A. A. A. El-Latif, "A secure federated learning framework for 5G networks," *IEEE Wireless Commun.*, vol. 27, no. 4, pp. 24–31, Aug. 2020.
- [32] B. Li, Y. Wu, J. Song, R. Lu, T. Li, and L. Zhao, "DeepFed: Federated deep learning for intrusion detection in industrial cyber-physical systems," *IEEE Trans. Ind. Informat.*, vol. 17, no. 8, pp. 5615–5624, Aug. 2021.
- [33] X. Wang, Y. Han, V. C. M. Leung, D. Niyato, X. Yan, and X. Chen, "Convergence of edge computing and deep learning: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 2, pp. 869–904, 2nd Quart., 2020.
- [34] X. Qiu, W. Zhang, W. Chen, and Z. Zheng, "Distributed and collective deep reinforcement learning for computation offloading: A practical perspective," *IEEE Trans. Parallel Distrib. Syst.*, vol. 32, no. 5, pp. 1085–1101, May 2021.
- [35] Z. Li, M. Xu, J. Nie, J. Kang, W. Chen, and S. Xie, "NOMA-enabled cooperative computation offloading for blockchain-empowered Internet of Things: A learning approach," *IEEE Internet Things J.*, vol. 8, no. 4, pp. 2364–2378, Feb. 2021.
- [36] W. Chen, X. Qiu, T. Cai, H.-N. Dai, Z. Zheng, and Y. Zhang, "Deep reinforcement learning for Internet of Things: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, early access, Apr. 13, 2021, doi: 10.1109/COMST.2021.3073036.
- [37] Y. Zhan, P. Li, Z. Qu, D. Zeng, and S. Guo, "A learning-based incentive mechanism for federated learning," *IEEE Internet Things J.*, vol. 7, no. 7, pp. 6360–6368, Jul. 2020.
- [38] Y. Zhan and J. Zhang, "An incentive mechanism design for efficient edge learning by deep reinforcement learning approach," in *Proc. IEEE INFOCOM Conf. Comput. Commun.*, Toronto, ON, Canada, 2020, pp. 2489–2498.
- [39] Y. Zhan, P. Li, and S. Guo, "Experience-driven computational resource allocation of federated learning by deep reinforcement learning," in *Proc. IEEE Int. Parallel Distrib. Process. Symp. (IPDPS)*, New Orleans, LA, USA, 2020, pp. 234–243.
- [40] S. Yu, X. Chen, Z. Zhou, X. Gong, and D. Wu, "When deep reinforcement learning meets federated learning: Intelligent multitimescale resource management for multiaccess edge computing in 5G ultra-dense network," *IEEE Internet Things J.*, vol. 8, no. 4, pp. 2238–2251, Feb. 2021.
- [41] Y. Zhan, J. Zhang, P. Li, and Y. Xia, "Crowdtraining: Architecture and incentive mechanism for deep learning training in the Internet of Things," *IEEE Netw.*, vol. 33, no. 5, pp. 89–95, Sep./Oct. 2019.
- [42] D. Kwon, J. Jeon, S. Park, J. Kim, and S. Cho, "Multiagent DDPG-based deep learning for smart ocean federated learning IoT networks," *IEEE Internet Things J.*, vol. 7, no. 10, pp. 9895–9903, Oct. 2020.
- [43] S. Messaoud, A. Bradai, O. Ben Ahmed, P. Quang, M. Atri, and M. S. Hossain, "Deep federated Q-learning-based network slicing for industrial IoT," *IEEE Trans. Ind. Informat.*, vol. 17, no. 8, pp. 5572–5582, Aug. 2021.
- [44] S. Lee and D.-H. Choi, "Federated reinforcement learning for energy management of multiple smart homes with distributed energy resources," *IEEE Trans. Ind. Informat.*, early access, Nov. 3, 2020, doi: 10.1109/TII.2020.3035451.
- [45] X. Wang, R. Li, C. Wang, X. Li, T. Taleb, and V. C. M. Leung, "Attention-weighted federated deep reinforcement learning for device-to-device assisted heterogeneous collaborative edge caching," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 1, pp. 154–169, Jan. 2021.
- [46] L. Lv, H. Jiang, Z. Ding, L. Yang, and J. Chen, "Secrecy-enhancing design for cooperative downlink and uplink NOMA with an untrusted relay," *IEEE Trans. Commun.*, vol. 68, no. 3, pp. 1698–1715, Mar. 2020.
- [47] Z. Ding, P. Fan, and H. V. Poor, "Impact of user pairing on 5g nonorthogonal multiple-access downlink transmissions," *IEEE Trans. Veh. Technol.*, vol. 65, no. 8, pp. 6010–6023, Aug. 2016.
- [48] Y. Cao *et al.*, "Power optimization for enhancing secrecy of cooperative user relaying noma networks," *IEEE Trans. Veh. Technol.*, vol. 69, no. 7, pp. 8008–8012, Jul. 2020.
- [49] S. Feng, D. Niyato, P. Wang, D. I. Kim, and Y.-C. Liang, "Joint service pricing and cooperative relay communication for federated learning," in *Proc. Int. Conf. Internet Things (iThings) IEEE Green Comput. Commun. (GreenCom) IEEE Cyber Phys. Soc. Comput. (CPSCom) IEEE Smart Data (SmartData)*, Atlanta, GA, USA, 2019, pp. 815–820.
- [50] S. R. Pandey, N. H. Tran, M. Bennis, Y. K. Tun, A. Manzoor, and C. S. Hong, "A crowdsourcing framework for on-device federated learning," *IEEE Trans. Wireless Commun.*, vol. 19, no. 5, pp. 3241–3256, May 2020.
- [51] Z. Yang, K. Liu, Y. Chen, W. Chen, and M. Tang, "Two-level Stackelberg game for IoT computational resource trading mechanism: A smart contract approach," *IEEE Trans. Services Comput.*, early access, Sep. 18, 2020, doi: 10.1109/TSC.2020.3024729.
- [52] J. Konečný, H. B. McMahan, D. Ramage, and P. Richtárik, "Federated optimization: Distributed machine learning for on-device intelligence," 2016. [Online]. Available: arXiv:1610.02527.
- [53] J. Schulman, P. Moritz, S. Levine, M. I. Jordan, and P. Abbeel, "High-dimensional continuous control using generalized advantage estimation," 2015. [Online]. Available: arXiv:1506.02438.
- [54] X. Wang, C. Wang, X. Li, V. C. M. Leung, and T. Taleb, "Federated deep reinforcement learning for Internet of Things with decentralized cooperative edge caching," *IEEE Internet Things J.*, vol. 7, no. 10, pp. 9441–9455, Oct. 2020.
- [55] V. Mnih *et al.*, "Asynchronous methods for deep reinforcement learning," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 1928–1937.



Minrui Xu is currently pursuing the B.S. degree with the School of Computer Science and Engineering, Sun Yat-sen University, Guangzhou, China.

His research interests include deep reinforcement learning and federated learning.



Jialiang Peng received the B.S. and M.S. degrees in computer science from Heilongjiang University, Harbin, China, in 2002 and 2005, respectively, and the Ph.D. degree in computer science from Harbin Institute of Technology, Harbin, in 2014.

He is an Associate Professor with the School of Data Science and Technology, Heilongjiang University. He worked as a Postdoctoral Scholar with the Norwegian University of Science and Technology, Gjøvik, Norway. His research interests include biometric recognition, data management, and cyberspace security.



B. B. Gupta (Senior Member, IEEE) received the Ph.D. degree in information and cyber security from the Indian Institute of Technology Roorkee, Roorkee, India, in 2009.

He was a Visiting Researcher with Yamaguchi University, Yamaguchi, Japan; Deakin University, Geelong, VIC, Australia; and the Swinburne University of Technology, Melbourne, VIC, Australia, in 2015, 2017, and 2018, respectively. He was a Visiting Professor with Temple University, Philadelphia, PA, USA, and Staffordshire University, Stoke-on-Trent, U.K., in June 2019 and July 2019, respectively. He is currently working as an Assistant Professor with the Department of Computer Engineering, National Institute of Technology Kurukshetra, Kurukshetra, India. He has published more than 250 research papers in international journals and conferences of high repute. His biography was selected and published in the 30th Edition of Marquis Who's Who in the World, 2012. His research interests include information security, cyber security, cloud computing, Web security, intrusion detection, and phishing.

Dr. Gupta received the Young Faculty Research Fellowship Award from MeitY, Government of India in 2017. He is working as a principal investigator of various research and development projects. He is serving as an Associate Editor of IEEE ACCESS and *International Journal of Internet and Computer Security* (Inderscience) and an Executive Editor of *International Journal of Information Technology, Control and Automation* (Inderscience).



Jiawen Kang received the M.S. and Ph.D. degrees from Guangdong University of Technology, Guangzhou, China, in 2015 and 2018, respectively.

He is currently a Postdoctoral Fellow with Nanyang Technological University, Singapore. His research interests mainly include blockchain and security and privacy protection in wireless communications and networking.



Zehui Xiong received the Ph.D. degree from Nanyang Technological University, Singapore, in 2020.

He is an Assistant Professor with the Pillar of Information Systems Technology and Design, Singapore University of Technology and Design, Singapore. Prior to that, he was a Researcher with Alibaba-NTU Joint Research Institute, Singapore. He was a Visiting Scholar with Princeton University, Princeton, NJ, USA, and the University of Waterloo, Waterloo, ON, Canada. He has published more than 90 research papers in leading journals and flagship conferences and four of them are ESI Highly Cited Papers. His research interests include wireless communications, network games and economics, blockchain, and edge intelligence.

Dr. Xiong was a recipient of the Chinese Government Award for Outstanding Students Abroad in 2019 and the NTU SCSE Best Ph.D. Thesis Runner-Up Award in 2020. He has won five best paper awards in international conferences and technical committee. He is currently serving as an editor or guest editor for many leading journals including IEEE transactions.



Zhenni Li (Member, IEEE) received the B.Sc. degree from the School of Physical Science and Electronics, Shanxi Datong University, Datong, China, in 2009, the M.Sc. degree from the School of Physics and Optoelectronic, Dalian University of Technology, Dalian, China, in 2012, and the Ph.D. degree from the School of Computer Science and Engineering, University of Aizu, Aizuwakamatsu, Japan, in 2015.

She is currently an Associate Professor with the School of Automation, Guangdong University of Technology, Guangzhou, China. Her research interests include machine learning, sparse representation, and deep learning.



Ahmed A. Abd El-Latif (Senior Member, IEEE) received the B.Sc. degree (Hons.) in mathematics and computer science and the M.Sc. degree in computer science from Menoufia University, Shebin El-Koom, Egypt, in 2005 and 2010, respectively, and the Ph.D. degree in computer science and technology from Harbin Institute of Technology, Harbin, China, in 2013.

He is an Associate Professor of Computer Science with Menoufia University and the School of Information Technology and Computer Science, Nile University, Giza, Egypt. He has authored and coauthored more than 140 papers, including refereed IEEE, ACM, Springer, and Elsevier journals, conference papers, books, and book chapters. His areas of interests are multimedia content encryption, secure wireless communication, IoT, applied cryptanalysis, perceptual cryptography, secret media sharing, information hiding, biometrics, forensic analysis in digital images, and quantum information processing.

Dr. El-Latif received many awards, the State Encouragement Award in Engineering Sciences in 2016, Arab Republic of Egypt; the Best Ph.D. Student Award from Harbin Institute of Technology, China, in 2013; and the Young Scientific Award, Menoufia University, Egypt, in 2014. Furthermore, he has been reviewing papers for 120+ international journals, including *IEEE Communications Magazine*, *IEEE INTERNET OF THINGS JOURNAL*, *Information Sciences*, *IEEE TRANSACTIONS ON NETWORK AND SERVICE MANAGEMENT*, *IEEE TRANSACTIONS ON SERVICES COMPUTING*, *Scientific Reports* (Nature), *Journal of Network and Computer Applications*, *Signal processing*, *Cryptologia*, *Journal of Network and Systems Management*, *Visual Communication and Image Representation*, *Neurocomputing*, and *Future Generation Computer Systems*. He is an Associate Editor of *Mathematical Problems in Engineering*, *Journal of Cyber Security and Mobility*, and *IET Quantum Communication*. He has many collaborative scientific activities with international teams in different research projects. He leading many special issues in several SCI and EI journals. He is a Fellow of the Academy of Scientific Research and Technology, Egypt. He is a member of ACM.